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RESEARCH

Myrtenol ameliorates inflammatory, oxidative, apoptotic, and hyperplasic effects of urethane-induced atypical adenomatous hyperplasia in the rat lung

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Abstract

Lung atypical adenomatous hyperplasia (AAH) is a forerunner of pulmonary adenocarcinoma. The drugs being utilized in the remediation of this type of hyperplasia have some adverse impacts. The present research focused on the potential anti-hyperplasia effect of myrtenol, an herbal terpenoid, on urethane-induced lung AAH in rats. Rats were injected with urethane (1.5 g/kg) thrice at 48 h intervals, and 20 weeks later, the animals were treated with 50 mg/kg myrtenol intraperitoneally once a day for 1 week. The ELISA method was used to measure inflammatory cytokines and oxidative parameters in the lung tissue and bronchoalveolar lavage fluid (BALF). The expression of NFκB and apoptotic/antiapoptotic factors (P53/Bcl-2) was evaluated by western blot and immunohistochemistry, respectively. H&E staining was performed for histopathological investigation. Histopathology confirmed the anti-hyperplasia effect of myrtenol, which was evidenced by the reduction of bronchoalveolar wall thickness and inflammation score. It also decreased hyperplasia progression by reducing Bcl-2, IL-10, p53, and Ki67. Compared with the urethane group, myrtenol normalized the activity of the oxidative stress markers malondialdehyde (MDA), total antioxidant capacity (TAC), glutathione peroxidase (GPX), and superoxide dismutase (SOD). Moreover, it showed an anti-inflammatory effect by decreasing lung and BALF IL-1β levels and NFκB expression. Myrtenol may have a promising effect on lung cancer treatment by counteracting lung hyperplasia via modulation of inflammation, oxidative stress, and apoptosis.

Keywords Urethane · Lung hyperplasia · Myrtenol · Inflammation · Oxidative stress · Apoptosis

Introduction

Development of bronchioloalveolar lung carcinoma appears to be stepwise, and the earliest lesion to be recognized is atypical adenomatous hyperplasia (AAH) (Kitamura et al. 1999). Lung AAH is a malignancy in both genders (Kalkanis et al. 2022) and a putative precursor lesion of adenocarcinoma of the lung (Mori et al. 2001). Lung AAH can be caused by a variety of factors, such as tobacco use and exposure to secondhand smoke,

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environmental tobacco smoke, occupational asbestos, and dust. Also, family history of the disease, genetics, age, race and ethnicity, diet, and infection are other risk factors (Cruz et al. 2011). Inflammation and lung AAH have been proven to have a strong link, and the significance of inflammation and the inflammatory microenvironment has received significant attention in lung hyperplasia research (Tan et al. 2021). It is well recognized that interleukin 1 beta (IL-1β) encourages the development, growth, and spread of tumors and chronic inflammation (Garon et al. 2020). Nuclear factor kappa B (NFκB) in myeloid cells primarily fosters lung AAH by mediating the release of inflammatory cytokines to create an inflammatory milieu conducive to hyperplasia growth (Chen et al. 2011). IL-10, an acid-sensitive homodimeric cytokine, is a pleiotropic molecule that controls autoimmunity, cell proliferation, survival, apoptosis, and angiogenesis and has anti-inflammatory properties (Vahl et al. 2017). Patients with lung AAH with IL-10 expression in their hyperplasia cells have a worse prognosis than those without (Zeni et al. 2007). Oxidative stress is a condition where there is an imbalance between the amount of reactive oxygen species (ROS) and a biological system's ability to quickly detoxify the reactive intermediates (Singh et al. 2014). Several studies have also shown a direct connection between systemic and organ redox imbalance and the development of lung hyperplasia (Zabłocka-Słowińska et al. 2018, 2019). Almost all malignancies harbor mutations in the p53 tumor suppressor gene, which is present in about 50% of all lung cancer cases (Ohno et al. 2001; Mattioni et al. 2015). p53, which is linked to the generation of ROS, is frequently mutated, and in this form, it accumulates in the cytoplasm and serves as an oncogene (Park et al. 2009). On the other hand, in lung AAH, there is an elevated expression of the antiapoptotic protein B cell lymphoma 2 (Bcl-2) (Zhu et al. 2006). Lung hyperplasia cells that express Bcl-2 and have mutant p53 may be highly resistant to the anticancer drug cisplatin and have limited susceptibility to apoptosis (Alam et al. 2022). Although p53 is known as a pro-apoptotic factor, many studies show that its expression can be increased in some cancers (Mori et al. 2001; Ohno et al. 2001; Vaughan et al. 2014, 2021). Moreover, it has been shown that the aberrant expression of p53 and Bcl-2 strongly correlates with and impacts survival and prognosis in patients with lung adenocarcinomas (Poposka et al. 2012). As a result, the balance of pro-apoptotic and antiapoptotic factors plays a vital role in cancer development (Borner et al. 1999). This study measured p53 as a pro-apoptotic factor and Bcl-2 as an antiapoptotic factor. However, the study unexpectedly ended with observing an increased level of p53, consistent with the findings of Mori et al., Ohno et al., and Vaughan et al. Alternatively, Ki67, a non-histone nuclear protein expressed at all phases of cell proliferation except the G0, was assessed and used as a tumor proliferation marker (Mehralikhani et al. 2020).

Studies have revealed that monoterpenes exhibit biological activities such as antioxidant, anti-inflammatory, chemotherapeutic, and chemo-preventive properties (17). *Myrtus communis* is a well-known traditional medicinal herb used to treat many illnesses worldwide (Rajizadeh et al. 2019, 2023a). One of the most potent components of *Myrtus communis* is myrtenol, a member of the terpene family, with anti-inflammatory (Oliveira et al. 2022), antioxidant, and anti-aging properties (Mrabti et al. 2023). We have demonstrated myrtenol's anti-inflammatory and anti-oxidative effects in experimental asthma (Esmaeilpour et al. 2023; Rajizadeh et al. 2023b).

This study employed urethane (also known as ethyl carbamate) to create lung hyperplasia. In the living organism, urethane is converted into vinyl carbamate epoxide intermediate compounds during a chemical reaction catalyzed by the CYP2E1 enzyme. These compounds attack DNA and cause a series of genetic alterations, leading to the development of lung tumors (Sozio et al. 2021).

Some synthetic drugs, such as erlotinib, rociletinib, and nivolumab, and new chemotherapy strategies for the treatment and prevention of hyperplasia have been developed (Zhao and Adjei 2015). However, these drugs have some disadvantages, such as vomiting, diarrhea, fever, and epigastric pain (Zhao and Adjei 2015; Muthu et al. 2019). Therefore, researchers are interested in developing new effective agents with fewer side effects.

The current study sought to determine the effect of myrtenol on the lungs of rats with urethane-induced AAH. In this regard, along with histopathology assessment, the level of oxidants, antioxidants, some pro-inflammatory and anti-inflammatory cytokines, and some indices of apoptosis/proliferation (Bcl-2, p53, and Ki67) were also assessed in the lungs.

Materials and method

Animals

Thirty-five Wistar albino rats weighing 200–250 g were used in the investigation. The rats were purchased from the Kerman University of Medical Sciences (KUMS) animal farm. They were kept at 22 ± 2 °C and 50–60% humidity with a 12-h light/dark cycle and unlimited access to regular diet and water. The KUMS Institutional Animal Care and Use Committee approved the procedures (Ethical code: IR.KMU.REC.1400.302). The rats were divided into five ($n = 7$ each) groups of CTL: healthy rats with no intervention, urethane: rats that received urethane, urethane/vehicle: rats that received urethane and DMSO 0.05% as myrtenol solvent, urethane/Myr: rats that received urethane and myrtenol, and urethane/CP: rats that received urethane and cisplatin as a gold standard drug.

Lung AAH induction and treatment protocol

For the induction of lung AAH, the rats were given intraperitoneal urethane (Sigma Aldrich) (1.5 g/kg) thrice at 48 h intervals (Parashar et al. 2022). Twenty weeks after the first urethane injection, treatment was initiated with intraperitoneal myrtenol (50 mg/kg) for 7 consecutive days (Rajizadeh et al. 2019).

BALF collection

Rats were sacrificed via anesthesia drug overdose (ketamine 80 mg/kg and xylazine 50 mg/kg), and BALF collection was done from the left lung. 2.5 mL of normal saline was instilled into the bronchoalveolar space by a catheter. The aspirated fluid is BALF (Khoramipour et al. 2023).

Evaluation of lung hyperplasia

The lungs and airways were immersed in 10% formalin. Then, slices were sectioned (using a microtome, Slee Co., Germany). Hematoxylin and eosin (H&E) staining was used to assess the histological alterations for general histopathology. A semi-quantified inflammatory score was utilized based on a previous study (Ibrahim et al. 2023). Also, the bronchiolar wall thickness (μm) and alveolar diameter (μm) were evaluated. The severity of lung injury was graded according to the study of Bánfi et al. (2009).

Immunohistochemistry

Lung tissue sections were immunostained with antibodies specific to p53 (USA, ScyTek, A00109), Bcl-2 (USA, ScyTek, A00119), and Ki67 (USA, ScyTek, A00095) (NeoMarkers: 1:100 dilutions). The specificity of the immunostaining was confirmed by control experiments that eliminated the primary antibody from the staining. Expression levels of p53 and Bcl-2 proteins were estimated by counting immunostained cells and measuring staining intensity according to methods previously reported. Briefly, score 0 indicates a percentage of positive cells less than 5%; score 1, 5–25%; score 2, 25–50%; score 3, 50–75%; and score 4 more than 75% positive cells. The intensity of DAB staining was also defined as score 1 for light yellow, score 2 for dark yellow or yellow–brown, and score 3 for brown staining. The Ki67 data are presented as percentages of positive cells in the fields. Five visual fields were evaluated on each slide, and the average score was used for statistical analysis (Gu et al. 2013a; Olivares-Hernández et al. 2022). Two pathologists scored all slides in a blinded fashion via an Olympia microscope.

Western blot

Western blot analyses were performed in four stages: total protein extraction (RIPA buffer), SDS-PAGE-based protein separation, protein separation (polyvinylidene difluoride [PVDF]), and target protein detection (protein-enhanced chemiluminescence [ECL] system) as previously described. Proteins are identified using specific antibodies (Santa Cruz Biotechnology Company, Sc-136548, Sc-47778). Data were analyzed using ImageJ 1.53t (Rajizadeh et al. 2023a).

Biochemical measurements in BALF and lung tissue

Total protein was estimated using the Bradford method. The oxidative stress factors in lung tissue were measured by kits from Navand Salamat Co., Iran. The cytokine levels in lung tissue and BALF were measured by the ELISA method via related kits (Karmania Pars Gene Co., Iran).

Lung hyperplasia index

The lung index is an index of hyperplasia and is assessed by the following formula: Lung index (%) = (lung weight/body weight) × 100 (Ibrahim et al. 2023).

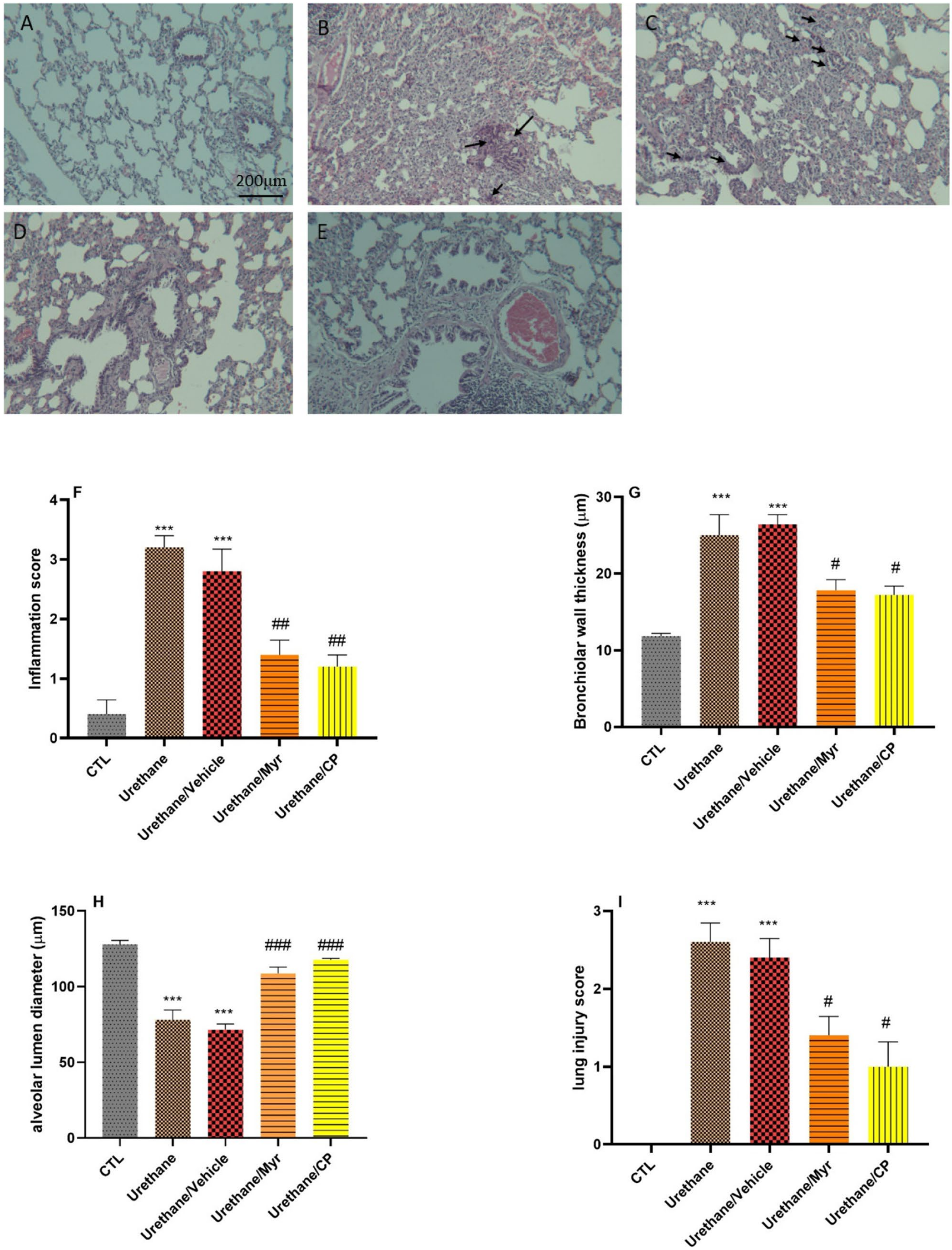
Statistical analysis

All statistical analyses were performed using version 8 of Prism software. One-way analysis of variance (ANOVA) was used to perform multi-group comparisons of means, and Tukey's post hoc test was used to compare the differences between groups. The result was considered statistically significant when the *P*-value was less than 0.05.

Results

Myrtenol improved histological lesions and reduced lung hyperplasia index

The results of H&E staining (Fig. 1) showed epithelial cell hyperplasia and atypical hyperplasia with mild to moderate atypia, elevated cell size, and nuclear-to-cytoplasmic ratio and hyperchromatic nuclei in the urethane groups while myrtenol ameliorated these lesions. In treatment groups, we observed mild hyperplasia without atypia and reduced inflammation and atelectasis. The inflammation score (Fig. 1F) and lung injury score (Fig. 1I) in the urethane


A
B
C
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200µm
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Inflammation score
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1
2
3
4
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Urethane
Urethane/Vehicle
Urethane/Myr
Urethane/CP
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◂**Fig. 1** Micrographs of the lung stained with H&E. **A** CTL (control); **B** urethane; **C** urethane/vehicle; **D** urethane/myrtenol; **E** urethane/CP. The graphs showed the effects of urethane and myrtenol on inflammation score (**F**), lung injury score (**I**), bronchiolar wall thickness (**G**), and alveolar diameter (**H**) in our study groups ($n=7$ in each group) (magnification, ×100; scale bar, 200 μm). Black arrows indicate hyperplasic cells. Data are presented as mean ± SEM. ***$P>0.001$ vs CTL. #$P>0.05$, ##$P>0.01$, ###$P>0.001$ *vs* urethane. CP, cisplatin

group significantly ($P<0.001$) elevated in comparison with the CTL group. Treatment with myrtenol significantly lowered the inflammation score ($P<0.01$) and lung injury score ($P<0.05$) in comparison with the urethane group. Bronchial wall thickness increased, and alveolar diameter decreased significantly in the urethane group ($P<0.001$), and treatment with myrtenol significantly ($P<0.05$) lowered bronchial wall thickness and increased alveolar diameter ($P<0.001$) in comparison with the urethane group. These impacts are comparable to the effects of cisplatin, which reduces hyperplasic cells.

Lung index was considerably elevated in the urethane group compared to the control group ($P<0.001$). The myrtenol group demonstrated a significant ($P>0.001$) reduction in the lung index in comparison with the urethane group (Fig. 2).

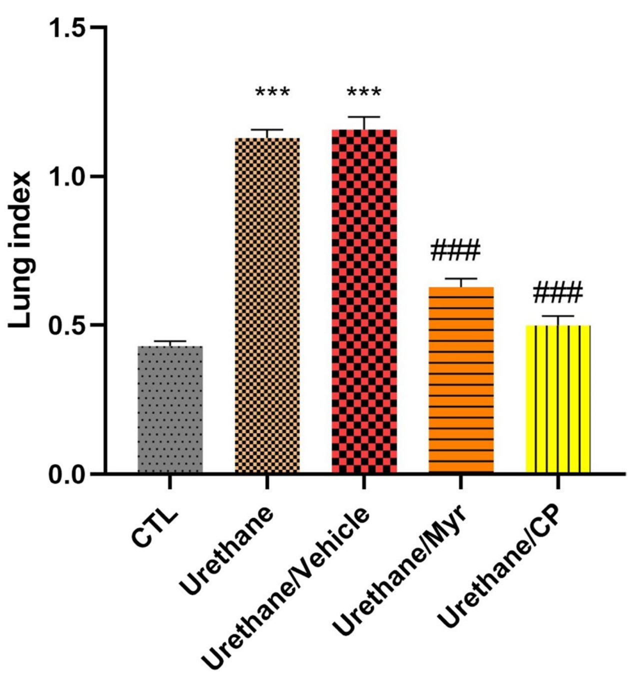

Fig. 2 Effect of myrtenol on lung index in urethane-induced lung hyperplasia in rats. ***$P>0.001$ *vs* CTL. ###$P>0.001$ *vs* urethane. CP, cisplatin

Myrtenol modulated the expression of p53, Bcl2, and Ki67

Figure 3 shows the micrographs of the immunohistochemical evaluation of Bcl-2 expression in lung tissue (A–E). The urethane group disclosed a significant ($P<0.001$) elevation in Bcl-2 expression compared to the CTL group. Treatment with myrtenol and cisplatin significantly ($P<0.01$) decreased Bcl-2 expression compared to the urethane groups.

Figure 4 illustrates the p53 expression in the study groups. A significant elevation was observed in the urethane group, especially in atypical hyperplastic areas compared to the CTL group. Curation with myrtenol significantly reduced p53 in comparison with the urethane group ($P<0.001$) (Fig. 4F). Myrtenol had similar effects as cisplatin in regulating p53 expression.

Figure 5 shows Ki67 expression among the groups. The expression of Ki67 was significantly increased in the urethane group compared to the control group ($P<0.001$). The administration of myrtenol and cisplatin significantly reduced Ki67 expression compared to the urethane group.

Myrtenol improved oxidative status in the lung tissue

Treatment of rats with urethane resulted in a significant ($P<0.001$) elevation in MDA levels (oxidant index) in untreated rats compared to their corresponding control groups (Fig. 6A). Administration of myrtenol to the rats with urethane-induced lung hyperplasia caused a significant ($P<0.001$) decrease in MDA levels.

In Fig. 6B, it is demonstrated that the TAC levels in the urethane group remarkably diminished compared to the CTL group ($P<0.001$). In animals that underwent treatment with myrtenol, the TAC level increased compared to the urethane group ($P<0.01$).

The impact of curation on antioxidant biomarkers (SOD and GPX) is illustrated in Fig. 6C, D; SOD and GPX activity considerably diminished in the urethane group compared to the CTL group ($P<0.001$). Myrtenol and cisplatin significantly recovered the SOD and GPX levels in comparison with the urethane group ($P<0.001$).

Myrtenol ameliorated inflammation in lung tissue and BALF

Figure 7 demonstrates the effect of myrtenol on IL-1β and IL-10 levels in the lung and BALF. Lung and BALF IL-1β levels were significantly elevated in the urethane group in comparison with the CTL group ($P<0.001$), and myrtenol

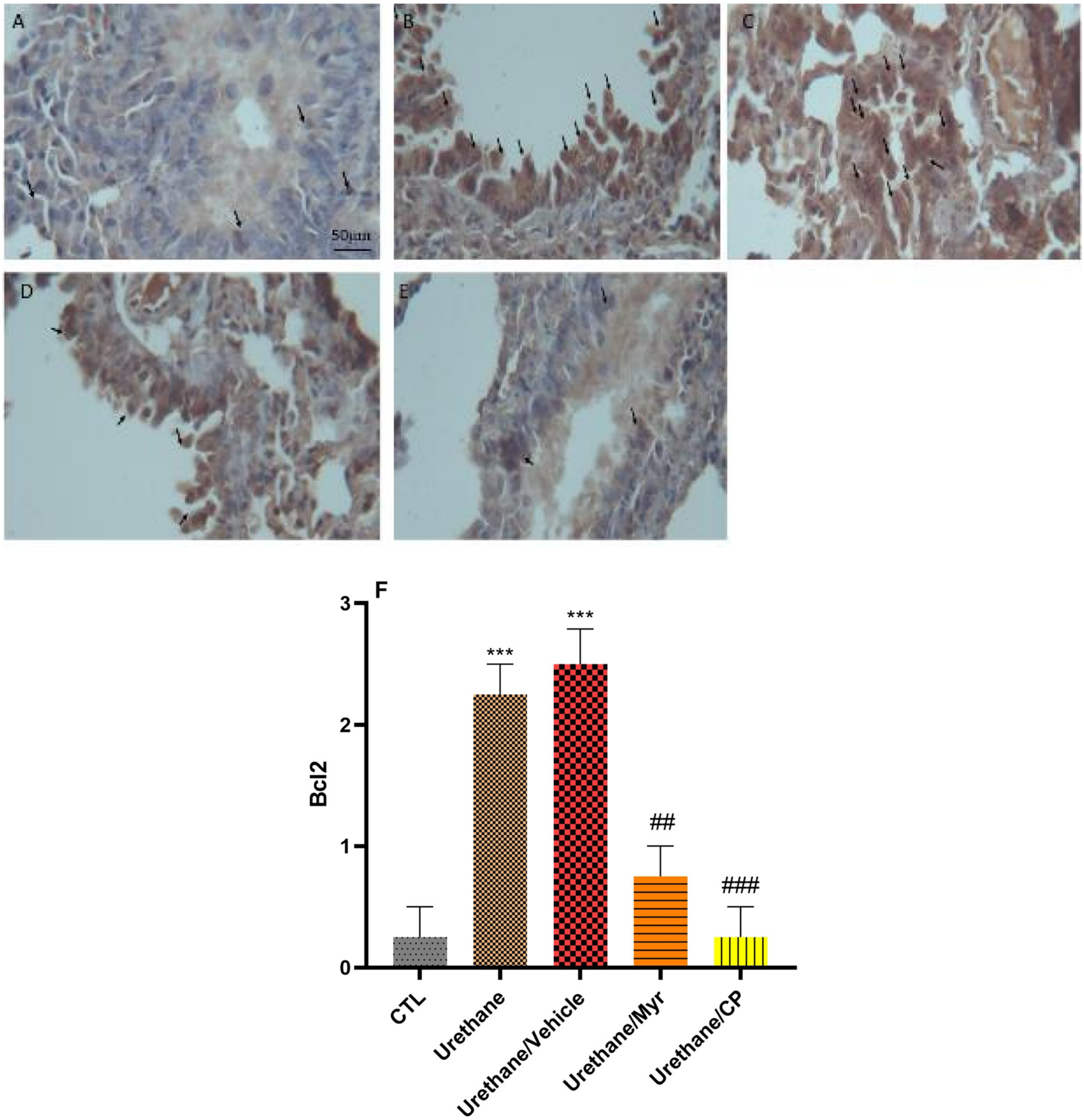

Fig. 3 Immunohistochemical assessment of the impacts of myrtenol on Bcl-2 expression (**A**–**F**) in lung tissue of rats (magnification, ×40; scale bar, 50 μm). **A** Control group, **B** urethane group, **C** urethane/vehicle group, **D** urethane/myrtenol, **E** urethane/CP group, **F** quantification of the Bcl-2 expression. Black arrows show Bcl-2-positive cells. Data are presented as mean ± SEM for $n = 7$ in each group ***$P < 0.001$ *vs* CTL. ###$P < 0.001$ *vs* urethane. CP, cisplatin

administration significantly recovered the IL-1β levels ($P < 0.001$). Also, IL-10 level was significantly ($P < 0.001$) higher in the lung and BALF of urethane groups compared to the control groups, and curation with myrtenol and cisplatin remarkably recovered the level of IL-10 in tissue and BALF ($P < 0.001$).

Myrtenol reduced NFκB expression in the lung tissue

Western blot analysis showed that urethane caused a significant ($P < 0.001$) elevation in the expression of NFκB in hyperplastic lungs compared to the control (Fig. 8). Administration of

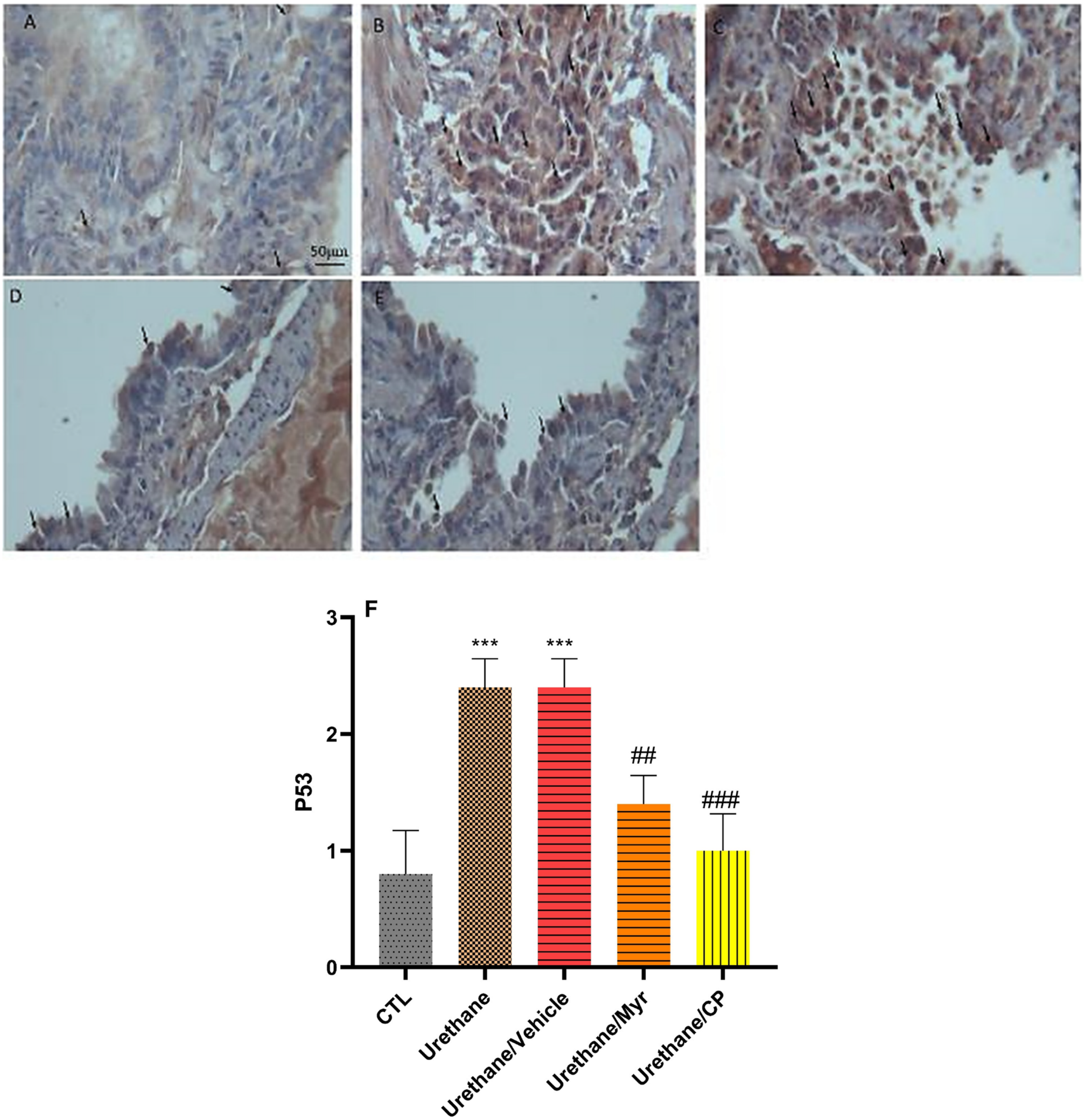

Fig. 4 Immunohistochemical assessment of the impact of myrtenol on p53 expression (**A**–**F**) in lung tissue of rats (magnification, ×40; scale bar, 50 µm). **A** Control group; **B** urethane group; **C** urethane/vehicle group, **D** urethane/myrtenol, **E** urethane/CP group, **F** quantification of the P53 expression. Black arrows show P53-positive cells. Data are presented as mean ± SEM for $n = 7$ in each group ***$P < 0.001$ *vs* CTL. ###$P < 0.001$ *vs* urethane. CP, cisplatin

myrtenol to the rats with lung hyperplasia significantly reduced NFκB expression compared to the urethane group ($P < 0.01$). No significant difference was found between the effect of myrtenol and cisplatin on NFκB expression (Fig. 8).

Discussion

This investigation was designed to evaluate the impacts of myrtenol on lung histopathology, BALF, tissue levels of cytokines, and lung markers of inflammation and oxidative

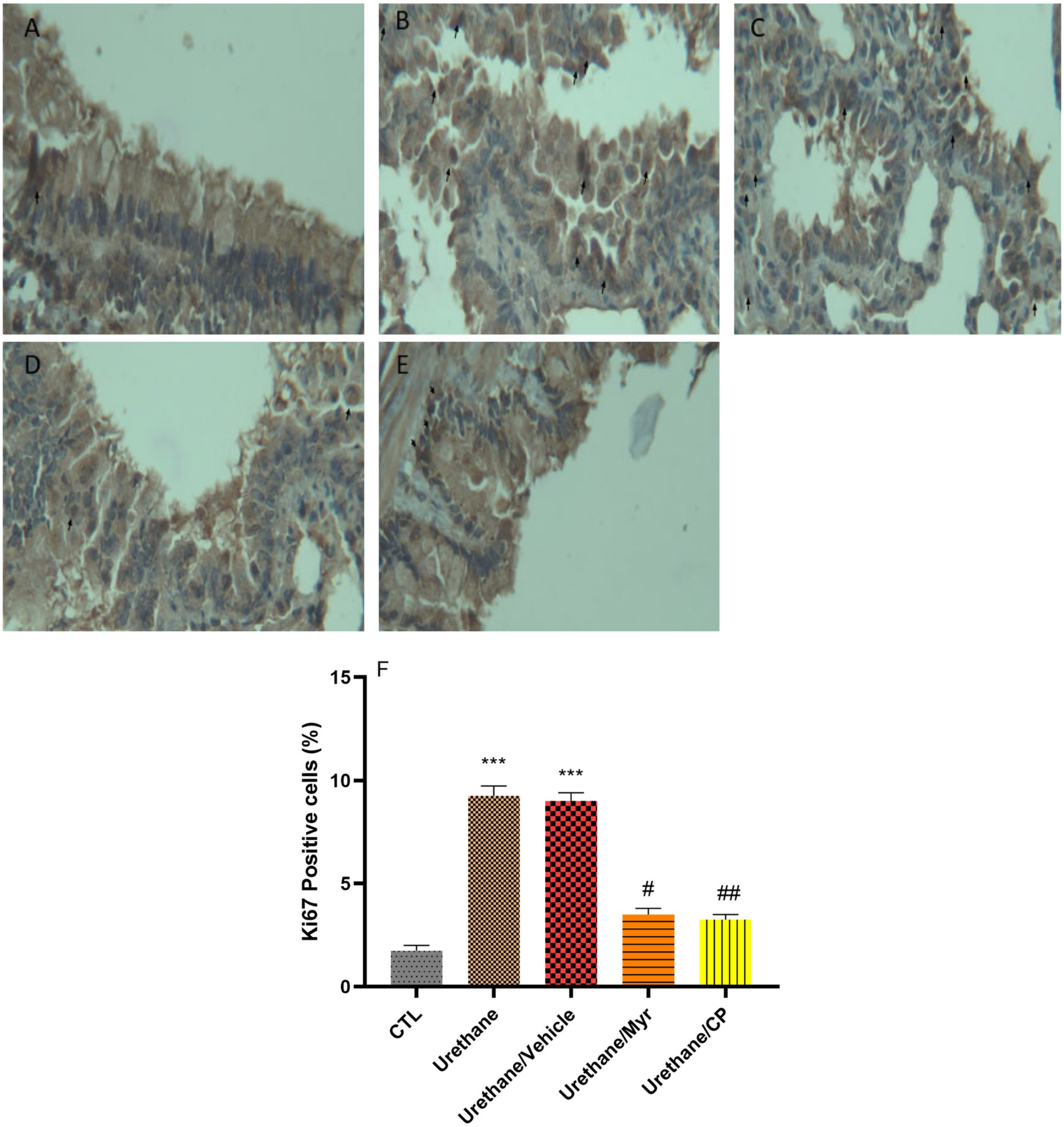

Fig. 5 Immunohistochemical assessment of the impact of myrtenol on Ki67 expression (**A**–**F**) in lung tissue of rats (magnification, ×40; scale bar, 50 µm). **A** Control group; **B** urethane group; **C** urethane/vehicle group, **D** urethane/myrtenol, **E** urethane/CP group, **F** quantification of the Ki67expression. Black arrows show ki67-positive cells. Data are presented as mean ± SEM for $n = 7$ in each group ***$P < 0.001$ *vs* CTL. #$P < 0.05$ and ##$P < 0.01$ *vs* urethane. CP, cisplatin

stress following urethane-induced hyperplasia. Our findings disclosed that myrtenol administration could alleviate lung injury by reducing lung injury and the levels of Bcl-2 and P53 and diminishing oxidative stress and inflammation.

It has been revealed that IL-1β elevation correlates with tumor development in various malignancies (Karki and Kanneganti 2019). The elevation of IL-1β has been found to increase inflammation-associated tumor invasiveness and stimulate the

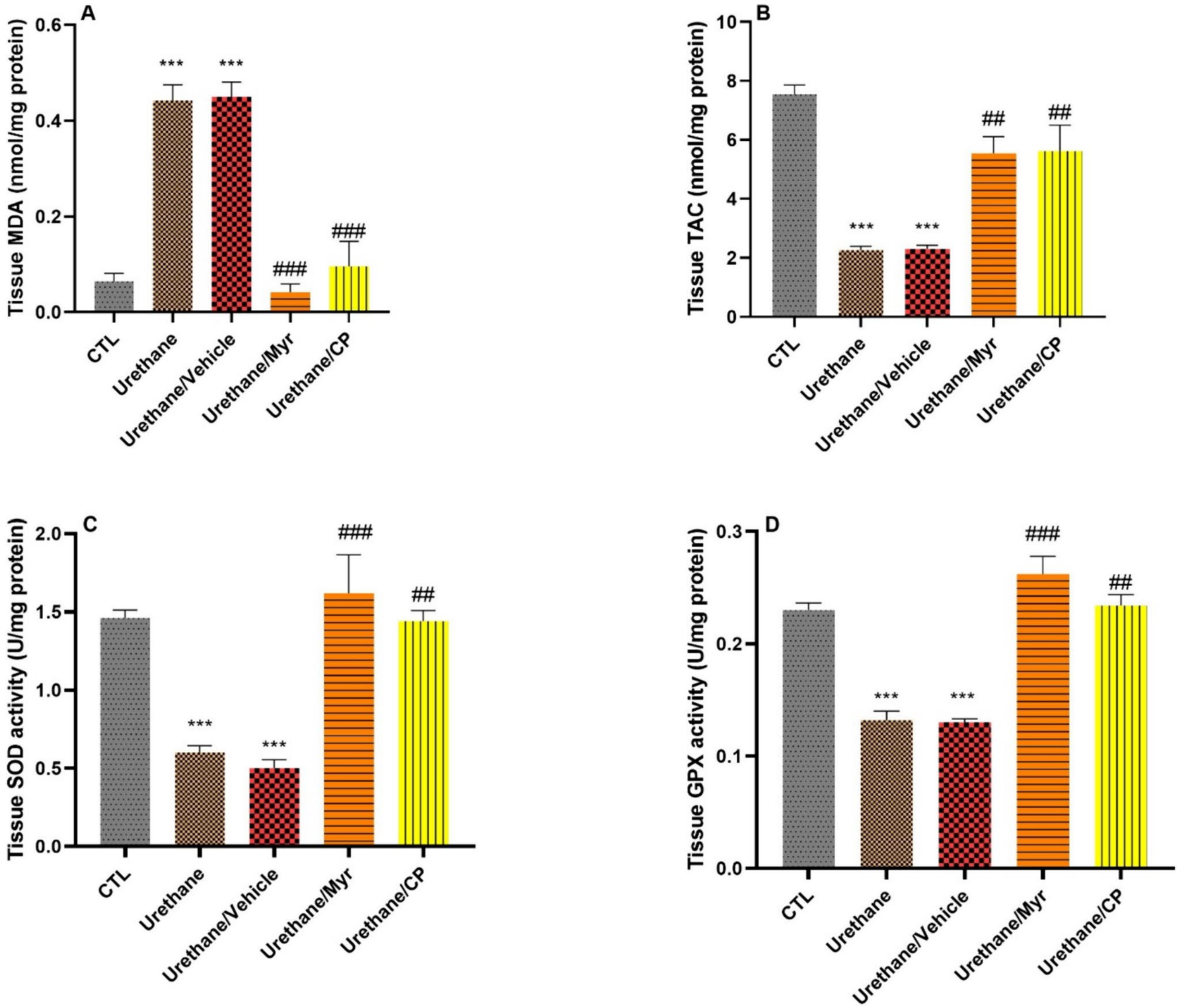

Fig. 6 Alterations in the oxidant and antioxidant enzymes in the lung tissue of the study groups. MDA (**A**), total antioxidant capacity (TAC) (**B**), superoxide dismutase activity (SOD) (**C**), glutathione peroxidase activity (GPx) (**D**). Data are presented as mean ± SEM ($n=7$ rats per group). ***$P<0.001$ *vs* CTL. ##$P<0.01$, ###$P>0.001$ *vs* urethane

tumor microenvironment to favor an increase in cell proliferation and angiogenesis (Multhoff et al. 2012; Karki and Kanneganti 2019). Inhibition of IL-1β signaling has been linked to suppressing tumor progression and enhancing tumor immunity (Apte et al. 2006; Garon et al. 2020). On the other hand, IL-1β can promote hyperplasia progression through NFκB activation (Diep et al. 2022). Although IL-10 is known as an anti-inflammatory cytokine, its role in hyperplasia is controversial. Several studies have shown that IL-10 is highly expressed in lung hyperplasia and mice models, which is associated with poor prognosis (Hsu et al. 2016; Vahl et al. 2017; Radwan et al. 2021). It seems that IL-10 promotes tumor development by suppressing the anti-tumor immune response (Radwan et al. 2021). In general, the immune-suppressive role of IL-10 resulted in tumor evasion from the immune system (Mirlekar 2022). In the present study, we observed a significant increase in IL-10 in the tissues and BALF of hyperplasia lungs, consistent with the findings of the above studies.

Oxidative stress is present in various hyperplasia cells. The redox imbalance is related to oncogenic stimulation (Valko et al. 2006). The increase in oxidative factors and decrease in antioxidants in the present study were associated with hyperplasia in the lungs of urethane groups, which were then restored toward normal levels by myrtenol treatment.

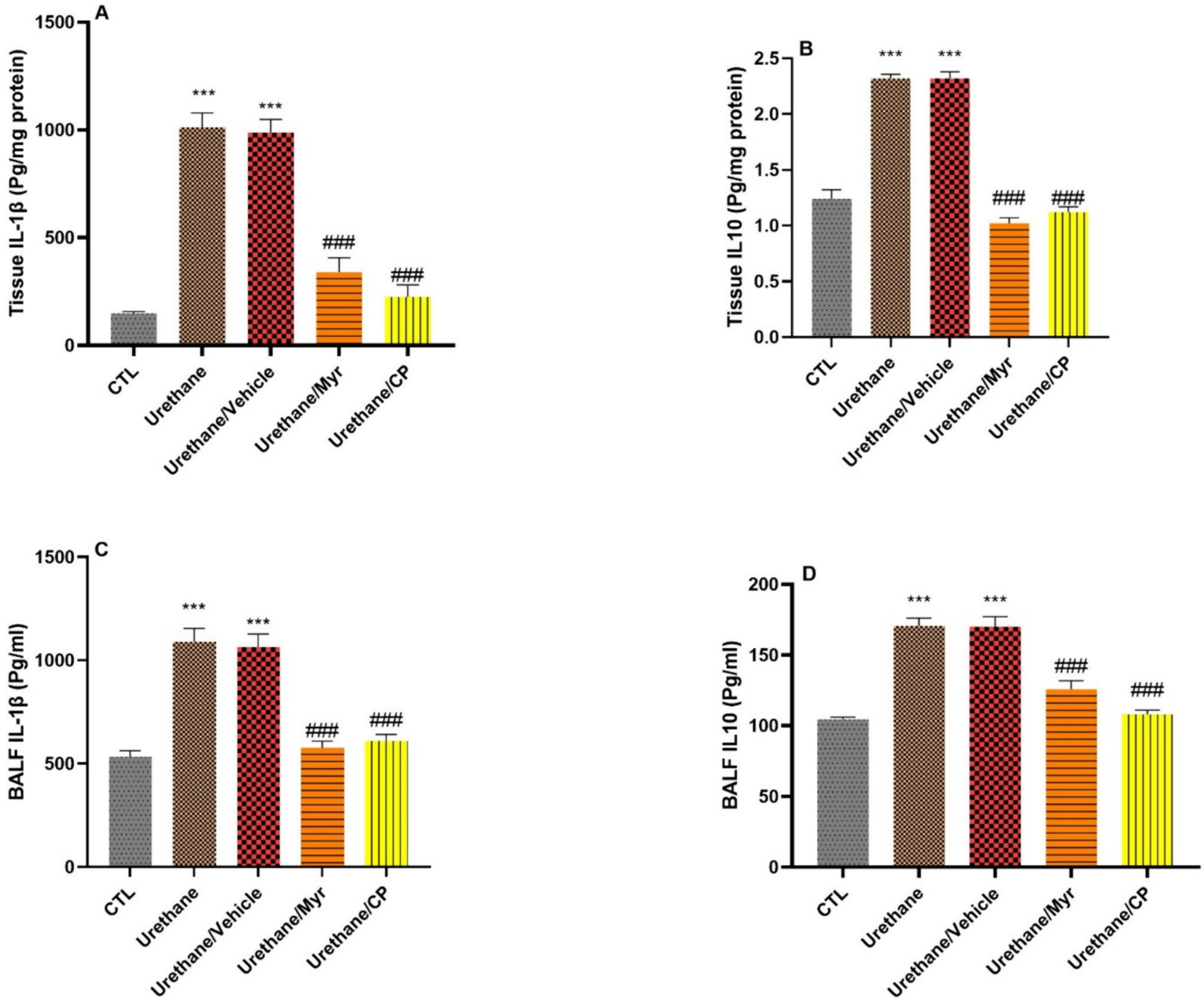

Fig. 7 ELISA (Enzyme-linked immunosorbent assay) analyses of IL-1β and IL-10 in the lung tissue (**A** and **B**) and BALF (**C** and **D**) of different groups. Urethane significantly increased the level of cytokines in the lung tissue and BALF. Treatment with myrtenol and cisplatin significantly compensated for the effects of urethane. Data are presented as mean ± SEM (n = 7 rat per group). BALF = Broncho alveolar lavage fluid. ***P < 0.001 vs CTL. ###P < 0.001 vs urethane

A considerable property of the hyperplasia cells is that they fail to undergo apoptosis, which in turn confers them a survival advantage over normal cells. Antiapoptotic protein, Bcl-2, is a protein that is highly upregulated in many hyperplasic lesions as compared to normal cells (Radha and Raghavan 2017). Bcl-2 suppresses the initiation of apoptosis by inhibiting mitochondrial permeability transition pores. Overexpression of Bcl-2 could inhibit cell apoptosis and increase the risk of malignant cell transformation (Gu et al. 2013a). The findings of the present study are consistent with the anti-hyperplasic effect of myrtenol. Based on our results, myrtenol's inhibition of Bcl-2 expression may have played a role in preventing epithelial cell lesions.

The tumor suppressor p53 is involved in cell growth regulation, cell cycle progression, DNA repair, and apoptosis, and mutations in the p53 gene, the most common genetic alterations in human cancers, can lead to the production of dysfunctional p53 proteins that unexpectedly allow the survival of genetically unstable cells that can turn into malignant cells. Mutant p53 proteins show a longer half-life than wild-type p53, resulting in accumulation in cancer cells. In lung cancer, p53 mutations happen early, and p53 overexpression is detected in preneoplastic lesions, such as bronchial dysplasia (Mattioni et al. 2015). p53 mutations can be divided into three groups (Vaughan et al. 2014, 2021): (1) loss of function, where p53 mutation leads to loss of its

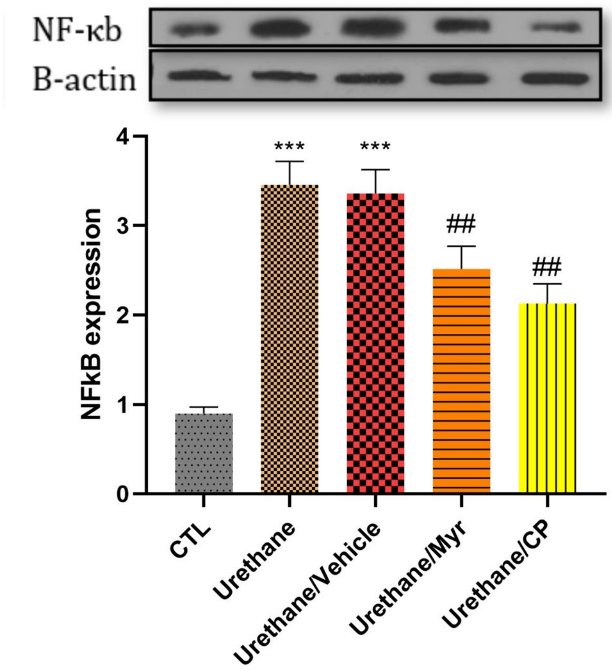

Fig. 8 Western blot analyses of NFκB in the lung tissue of study groups. Data are means ± SEM ($n = 5$ rats per group). ***$P < 0.001$ *vs* CTL. ##$P < 0.01$ *vs* urethane

tumor suppressor function; (2) dominant negative, where a mutant copy hetero-oligomerizes with a wild-type (WT) copy poisoning the WT tumor suppressor function; and (3) gain of function, where the cells acquire a new dominant oncogenic property leading to oncogenesis. p53 mutation following lung hyperplasia has been observed in some other studies (Wu et al. 2006; Gu et al. 2013b). Consistently, it has been shown that the downregulation of mutant p53 increases the expression of p53 target proteins and suppresses cancer cell growth (Zhu et al. 2013). It is probable that the downregulation of mutant p53 expression by myrtenol, at least partly, accounted for the prevention of lung hyperplasia.

Regarding the anti-inflammatory and anti-oxidative effects of myrtenol, the results of our study are consistent with this property of myrtenol. Also, concerning apoptosis, our results showed that myrtenol had a pro-apoptotic effect by reducing the antiapoptotic protein Bcl-2 and the tumorigenesis marker Ki67.

Conclusion

Overall, our findings demonstrated that myrtenol administration might induce anti-hyperplasia effects, and this effect can be attributable to the reduction of inflammation, oxidative stress, and modulation of hyperplasia-related genes such as Bcl-2, P53, and Ki67 (Graphical abstract). This may propose myrtenol as a target for more preclinical experiments toward providing a new drug against lung hyperplasia.

Limitations

The statement that p53 undergoes mutation following urethane administration was inferred from the results we provided in this study. As this is not the main context of the present study, we propose that additional tests, such as real-time PCR and sequencing, be performed to confirm the type and quality of this mutation in future studies.

Authors contributions Sedigheh Amiresmaili: writing—review and editing, writing—original draft, and supervision; Mohammad Amin Rajizadeh: methodology and investigation; Elham Jafari: methodology; Mohammad Abbas Bejeshk: methodology; Fouzieh Salimi: writing—review and editing; Amirhossein Moslemizadeh: writing—original draft and investigation; Hamid Najafipour: methodology, conceptualization, and supervision. The authors declare that all data were generated in-house and that no paper mill was used.

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Data availability Data will be provided by request from the corresponding author.

Declarations

Ethical approval The KUMS' Institutional Animal Care and Use Committee approved the procedures (Ethical code: IR.KMU.REC.1400.302).

Consent to participate N/A

Consent for publication N/A

Competing interests The authors declare there is no conflict of interest.

References

Alam M, Alam S, Shamsi A, Adnan M, Elasbali AM, Al-Soud WA, Alreshidi M, Hawsawi YM, Tippana A, Pasupuleti VR (2022) Bax/Bcl-2 cascade is regulated by the EGFR pathway: therapeutic targeting of non-small cell lung cancer. Front Oncol 12:933

Apte RN, Dotan S, Elkabets M, White MR, Reich E, Carmi Y, Song X, Dvozkin T, Krelin Y, Voronov E (2006) The involvement of IL-1 in tumorigenesis, tumor invasiveness, metastasis and tumor-host interactions. Cancer Metastasis Rev 25:387–408

Bánfi A, Tiszlavicz L, Székely E, Peták F, Tóth-Szüki V, Baráti L, Bari F, Novák Z (2009) Development of bronchus-associated lymphoid tissue hyperplasia following lipopolysaccharide-induced lung inflammation in rats. Exp Lung Res 35:186–197

Borner M, Brousset P, Pfanner-Meyer B, Bacchi M, Vonlanthen S, Hotz M, Altermatt H, Schlaifer D, Reed J, Betticher D (1999) Expression of apoptosis regulatory proteins of the Bcl-2 family and p53 in primary resected non-small-cell lung cancer. Br J Cancer 79:952–958

Chen W, Li Z, Bai L, Lin Y (2011) NF-kappaB, a mediator for lung carcinogenesis and a target for lung cancer prevention and therapy. Front Biosci 16:1172

Cruz CSD, Tanoue LT, Matthay RA (2011) Lung cancer: epidemiology, etiology, and prevention. Clin Chest Med 32:605–644

Diep S, Maddukuri M, Yamauchi S, Geshow G, Delk NA (2022) Interleukin-1 and nuclear factor kappa b signaling promote breast cancer progression and treatment resistance. Cells 11:1673

Esmaeilpour K, Jafari E, Rostamabadi F, Khaleghi M, Akhgarandouz F, Hosseini M, Najafipour H, Khodadoust M, Sheibani V, Rajizadeh MA (2023) Myrtenol inhalation mitigates asthma-induced cognitive impairments: an electrophysiological, behavioral, histological, and molecular study. Mol Neurobiol 61:4891–4907. https://doi.org/10.1007/s12035-023-03863-1

Garon EB, Chih-Hsin Yang J, Dubinett SM (2020) The role of interleukin 1β in the pathogenesis of lung cancer. JTO Clin Res Rep 1:100001

Gu Q, Hu C, Chen Q, Xia Y (2013a) Tea polyphenols prevent lung from preneoplastic lesions and effect p53 and bcl-2 gene expression in rat lung tissues. Int J Clin Exp Pathol 6:1523

Gu Q, Hu C, Chen Q, Xia Y (2013b) Tea polyphenols prevent lung from preneoplastic lesions and effect p53 and bcl-2 gene expression in rat lung tissues. Int J Clin Exp Pathol 6:1523–1531

Hsu TI, Wang YC, Hung CY, Yu CH, Su WC, Chang WC, Hung JJ (2016) Positive feedback regulation between IL10 and EGFR promotes lung cancer formation. Oncotarget 7:20840–20854

Ibrahim MA, Mohamed SR, Dkhil MA, Thagfan FA, Abdel-Gaber R, Soliman D (2023) The effect of Moringa oleifera leaf extracts against urethane-induced lung cancer in rat model. Environ Sci Pollut Res Int 30:37280–37294

Kalkanis A, Papadopoulos D, Testelmans D, Kopitopoulou A, Boeykens E, Wauters E (2022) Broncho-alveolar lavage fluid-isolated biomarkers for the diagnostic and prognostic assessment of lung cancer. Diagnostics 12:2949

Karki R, Kanneganti T-D (2019) Diverging inflammasome signals in tumorigenesis and potential targeting. Nat Rev Cancer 19:197–214

Khoramipour K, Bejeshk MA, Rajizadeh MA, Najafipour H, Dehghan P, Farahmand F (2023) High-intensity interval training ameliorates molecular changes in the hippocampus of male rats with the diabetic brain: the role of adiponectin. Mol Neurobiol 60:3486–3495

Kim C-H, Lee HS, Park J-H, Choi J-H, Jang S-H, Park Y-B, Lee MG, Hyun IG, Kim KI, Kim HS (2015) Prognostic role of p53 and Ki-67 immunohistochemical expression in patients with surgically resected lung adenocarcinoma: a retrospective study. J Thorac Dis 7:822

Kitamura H, Kameda Y, Ito T, Hayashi H (1999) Atypical adenomatous hyperplasia of the lung: implications for the pathogenesis of peripheral lung adenocarcinoma. Am J Clin Pathol 111:610–622

Mattioni M, Soddu S, Prodosmo A, Visca P, Conti S, Alessandrini G, Facciolo F, Strigari L (2015) Prognostic role of serum p53 antibodies in lung cancer. BMC Cancer 15:1–11

Mehralikhani A, Movahedi M, Larypoor M, Golab F (2020) The role of fennel seed extract on the expression pattern of dysadherin, E-cadherin and Ki67 in metastatic lung cancer in BALB/C female mice. Herb Med J 5. https://doi.org/10.22087/hmj.v5i4.832

Mirlekar B (2022) Tumor promoting roles of IL-10, TGF-β, IL-4, and IL-35: its implications in cancer immunotherapy. SAGE Open Med 10:20503121211069012

Mori M, Rao SK, Popper HH, Cagle PT, Fraire AE (2001) Atypical adenomatous hyperplasia of the lung: a probable forerunner in the development of adenocarcinoma of the lung. Mod Pathol 14:72–84

Mrabti HN, Jaouadi I, Zeouk I, Ghchime R, El Menyiy N, El Omari N, Balahbib A, Al-Mijalli SH, Abdallah EM, El-Shazly M (2023) Biological and pharmacological properties of myrtenol: a review. Curr Pharm Des 29:407–414

Multhoff G, Molls M, Radons J (2012) Chronic inflammation in cancer development. Front Immunol 2:98

Muthu V, Mylliemngap B, Prasad KT, Behera D, Singh N (2019) Adverse effects observed in lung cancer patients undergoing first-line chemotherapy and effectiveness of supportive care drugs in a resource-limited setting. Lung India: Off Organ Indian Chest Soc 36:32

Ohno J, Horio Y, Sekido Y, Hasegawa Y, Takahashi M, Nishizawa J-i, Saito H, Ishikawa F, Shimokata K (2001) Telomerase activation and p53 mutations in urethane-induced A/J mouse lung tumor development. Carcinogenesis 22:751–756

Olivares-Hernández A, del Barco ME, Miramontes-González JP, Figuero-Pérez L, Pérez-Belmonte L, Martín-Vallejo J, Martín-Gómez T, Escala-Cornejo R, Vidal-Tocino R, Hernández LB (2022) Immunohistochemical assessment of the P53 protein as a predictor of non-small cell lung cancer response to immunotherapy. Front Biosci Landmark 27:88

Oliveira JP, Abreu FF, Bispo JMM, Cerqueira AR, Santos JRd, Correa CB, Costa SK, Camargo EA (2022) Myrtenol reduces orofacial nociception and inflammation in mice through p38-MAPK and cytokine inhibition. Front Pharmacol 13:1874

Parashar P, Tripathi CB, Arya M, Kanoujia J, Singh M, Yadav A, Kumar A, Guleria A, Saraf SA (2022) Biotinylated naringenin intensified anticancer effect of gefitinib in urethaneinduced lung cancer in rats: favourable modulation of apoptotic regulators and serum metabolomics (Retraction of Vol 46, Pg S598, 2018). TAYLOR & FRANCIS LTD 2-4 PARK SQUARE, MILTON PARK, ABINGDON OR14 4RN, OXON …

Park HS, Kim SR, Lee YC (2009) Impact of oxidative stress on lung diseases. Respirology 14:27–38

Poposka BI, Smickoska S, Petrusevska G (2012) Immunohistochemical expression of Bcl-2 and p53 in patients with lung cancer: correlation with survival time. Eur Respiratory Soc

Radha G, Raghavan SC (2017) BCL2: a promising cancer therapeutic target. Biochimica et Biophysica Acta (BBA) Rev Cancer 1868:309–314

Radwan E, Ali M, Faied SM, Omar HM, Mohamed WS, Abd-Elghaffar SK, Sayed AA (2021) Novel therapeutic regimens for urethane-induced early lung cancer in rats: combined cisplatin nanoparticles with vitamin-D3. IUBMB Life 73:362–374

Rajizadeh MA, Najafipour H, Fekr MS, Rostamzadeh F, Jafari E, Bejeshk MA, Masoumi-Ardakani Y (2019) Anti-inflammatory and anti-oxidative effects of myrtenol in the rats with allergic asthma. Iran J Pharm Res IJPR 18:1488

Rajizadeh MA, Khaksari M, Bejeshk MA, Amirkhosravi L, Jafari E, Jamalpoor Z, Nezhadi A (2023a) The role of inhaled estradiol and myrtenol, alone and in combination, in modulating behavioral and functional outcomes following traumatic experimental brain injury: hemodynamic, molecular, histological and behavioral study. Neurocritical Care 1–21

Rajizadeh MA, Nematollahi MH, Jafari E, Bejeshk MA, Mehrabani M, Rostamzadeh F, Fekri MS, Najafipour H (2023b) Formulation and evaluation of the anti-inflammatory, anti-oxidative, and anti-remodelling effects of the niosomal myrtenol on the lungs of asthmatic rats. Iran J Allergy Asthma Immunol 22:265–280. https://doi.org/10.18502/ijaai.v22i3.13054

Singh Z, Karthigesu IP, Singh P, Rupinder K (2014) Use of malondialdehyde as a biomarker for assessing oxidative stress in different disease pathologies: a review. Iran J Public Health 43:7–16

Sozio F, Schioppa T, Sozzani S, Del Prete A (2021) Urethane-induced lung carcinogenesis. Methods Cell Biol 163:45–57. https://doi.org/10.1016/bs.mcb.2020.09.005

Stakišaitis D, Mozūraitė R, Kavaliauskaitė D, Šlekienė L, Balnytė I, Juodžiukynienė N, Valančiūtė A (2017) Sex-related differences of urethane and sodium valproate effects on Ki-67 expression in urethane-induced lung tumors of mice. Exp Ther Med 13:2741–2750

Tan Z, Xue H, Sun Y, Zhang C, Song Y, Qi Y (2021) The role of tumor inflammatory microenvironment in lung cancer. Front Pharmacol 12:688625

Thalappil MA, Singh P, Carcereri de Prati A, Sahoo SK, Mariotto S, Butturini E (2024) Essential oils and their nanoformulations for breast cancer therapy. Phytother Res 38:556–591

Vahl JM, Friedrich J, Mittler S, Trump S, Heim L, Kachler K, Balabko L, Fuhrich N, Geppert CI, Trufa DI, Sopel N, Rieker R, Sirbu H, Finotto S (2017) Interleukin-10-regulated tumour tolerance in non-small cell lung cancer. Br J Cancer 117:1644–1655

Valko M, Rhodes C, Moncol J, Izakovic M, Mazur M (2006) Free radicals, metals and antioxidants in oxidative stress-induced cancer. Chem Biol Interact 160:1–40

Vaughan CA, Singh S, Subler MA, Windle JJ, Inoue K, Fry EA, Pillappa R, Grossman SR, Windle B, Andrew Yeudall W (2021) The oncogenicity of tumor-derived mutant p53 is enhanced by the recruitment of PLK3. Nat Commun 12:704. https://doi.org/10.1038/s41467-021-20928-8

Vaughan C, Pearsall I, Yeudall A, Deb SP, Deb S (2014) p53: its mutations and their impact on transcription. Subcell Biochem 85:71–90. https://doi.org/10.1007/978-94-017-9211-0_4

Wu C-H, Lin H-H, Yan F-P, Wu C-H, Wang C-J (2006) Immunohistochemical detection of apoptotic proteins, p53/Bax and JNK/FasL cascade, in the lung of rats exposed to cigarette smoke. Arch Toxicol 80:328–336

Zabłocka-Słowińska K, Płaczkowska S, Prescha A, Pawełczyk K, Kosacka M, Porębska I, Grajeta H (2018) Systemic redox status in lung cancer patients is related to altered glucose metabolism. PLoS ONE 13:e0204173

Zabłocka-Słowińska K, Płaczkowska S, Skórska K, Prescha A, Pawełczyk K, Porębska I, Kosacka M, Grajeta H (2019) Oxidative stress in lung cancer patients is associated with altered serum markers of lipid metabolism. PLoS ONE 14:e0215246

Zeni E, Mazzetti L, Miotto D, Cascio NL, Maestrelli P, Querzoli P, Pedriali M, De Rosa E, Fabbri L, Mapp C (2007) Macrophage expression of interleukin-10 is a prognostic factor in nonsmall cell lung cancer. Eur Respir J 30:627–632

Zhao Y, Adjei AA (2015) New strategies to develop new medications for lung cancer and metastasis. Cancer Metastasis Rev 34:265–275

Zhu C, Shih W, Ling C, Tsao M (2006) Immunohistochemical markers of prognosis in non-small cell lung cancer: a review and proposal for a multiphase approach to marker evaluation. J Clin Pathol 59:790–800

Zhu H-B, Yang K, Xie Y-Q, Lin Y-W, Mao Q-Q, Xie L-P (2013) Silencing of mutant p53 by siRNA induces cell cycle arrest and apoptosis in human bladder cancer cells. World J Surg Oncol 11:1–11

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